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ORIGINAL RESEARCH

Sentimental analysis over twitter data using clustering based machine learning algorithm

Sharon Susan Jacob[1] · R. Vijayakumar[1]

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Abstract

Using Internet technologies, people often share their thoughts, feedback, news and information with others. Owing to social media sites, speed and ease of contact improved. User posts their views on public sentiment web sites like Facebook, Instagram, Twitter, blogs, WhatsApp, Snapchat, LinkedIn, etc. Thousands of posts, millions of tweets and thousands of letters are posted each day. Twitter is one of them that is now being widely popular in these social media sites. It offers an easy and quick way to evaluate the opinions of consumers on a product or service. The approach to be used to mark customer's impressions or opinions of a product is to establish an sentimental analysis system. Twitter is a microblogging platform where users can submit feedback to a community of followers in the form of ratings or tweets in bigdata. A tweet may be defined as positive, negative or neutral depending on the viewpoint shared. Here we investigate the sentiment of Twitter messages in this paper using the clustering approach based on machine learning (ML) algorithm. Our tests are carried out in a qualified and test collection composed of large data from tweets from one lakh of results, showing our work to determine if a tweet is positive or negative.

Keywords Twitter · Sentiment analysis · Clustering · Big data · Machine learning algorithm

1 Introduction

With an advent of Smartphone and I-phone, today's generation become very active in online social networking websites like Twitter, Facebook and Instagram. These websites becomes more and more powerful tool for the people from different culture, religion, background and interest for sharing views and thoughts. With an increasing popularity the online activity gradually increases, so does spread hateful activities. These hateful activities in turn exploits social infrastructure. Although these OSN websites provide an open space to public to discuss about their thoughts and opinion over different background, events, cultures, believes, their life via posting comments, blog posts, and exchanged of messages. But as this discussion is open and no one is physically involved so it is nearly impossible to control their emotions in term of the content (King and Sutton 2013). In addition, with different conditions, customs and belief, most of public make use of aggressive, violent and hateful speech while discussing with other people who do not contribute to the same backgrounds. Conversely, with the fast development of OSN, exchange of thoughts about any controversial topic or some big events convert into conflicts which in turn divide people into two categories, one supports it and one contradicting it (Peter 2002). In contrast to "clean speech", "hate speech" is considerably simpler to spread among youngsters as well as elder ones.

Hate speech or incitement to hatred is, in general terms, any act of communication that interiorizes or incites against a person or group, based on characteristics such as race, gender, ethnicity, nationality, religion, sexual orientation or other discriminating aspect. When a homosexual is offended by his sexual orientation, all homosexuals are offended, just as when a black man is offended for the simple reason that he is black, all blacks are offended. Hate speech goes beyond the limits of common sense, since it aims to promote violence, discrimination or prejudice to the detriment of a group or cluster of people (Chouchani and Abed 2020). This is the reason hate speech is viewed as one of major issue that numerous nations and associations have been standing facing (Singh et al. 2018). With the spread of web, and the

✉ Sharon Susan Jacob
sharonsusanjacob.phd@gmail.com

1 School of Computer Sciences, Mahatma Gandhi University, Kottayam, Kerala, India

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development of online social network, this issue turns out to be more serious, since the communication between individuals is not direct, and individuals' speech become more violent as they feel safe physically (Awais et al. 2019).

Leading to these hateful activities in social network and internet, tensions amongst the group of individual creates which in turn affects business (Sangeetha and Prabha 2020); economy and sometimes online conflicts will convert into real time conflicts (Edara et al. 2019). Considering to this, many websites like twitter, youtube, facebook etc. will put restriction on using hateful or offensive language. After putting out such restriction still it is very difficult to control and filtered out the content.

Developing effective method of detecting hateful speech is one of the foremost for Online social networking website. In this context OSN websites invest hundreds of millions of euros every year on this task (Björn and Sikdar 2017; Guardian 2017; Natasha 2017) and still being criticized for unsatisfactory result. This is because most of all these work identify and filter out hateful speech manually.This manual process is very time consuming, labor intensive and not scalable (Ying et al. 2012; Björn and Sikdar 2017; Zeerak and Hovy 2016). In recent years large number of researches carried out in way of automation but no one will work on real time twitter tweets.

Our approach performs detection of real time twitter tweets based on priority based PoS Dictionary tagging (Derczynski et al. 2013), sentiment sensitive priority dictionary tagging, and direct stated feature extraction pair generation.

This paper is organized in the following pattern. In Sect. 2 of the paper, we present motivation, and novelty of the work. In Sect. 3 we describe some of related work and explain novelty of our work. In Sect. 4 we describe our proposed work and its flow to detect hateful speech and how sentiment sensitive dictionary is created and features are extracted. Dataset description, and result analysis are depicted and described in Sect. 5. The concluding remarks and future prospects of the research are stated in Sect. 6.

2 Motivations and novelty of work

Hateful speech are the language that contain abusive language targeting to specific group (may be an organization, religion, a country etc.) or an individual (including any product, a celebrity or, a politician) or particular groups. Detecting hateful post and offensive speech is not only important for studying public sentiment over one community to another but also helpful in investigating hate crime rate and threats amongst people. In Europe, survey report focusing youth in the EEA (European Economic Area) region depicts that increase in hate speech and connected crimes is based on religious thoughts, background, gender or race or, as 80% of perpetrator encounter due to online hate speech and 40% felt threatened or attacked EEANews (2017). King and Sutton (2013) states that following 9/11 event, hate crime related to anti-Islamic motive will increase up to 481, 58% of them were commit in two weeks after the event. This will motivate the researchers to identify hate speech from tweet comments. It is not scalable to filter out hateful tweets manually; this will raise another opportunity to researchers and motivate them to identify some automated ways.

Many researchers perform detection in automated way Watanabe et al. (2018), Warner and Julia (2012), Paula and Nunes (2018), Njagi et al. (2015). These works usually use semantic based approach in which text contents are clustered into non-hate and hateful. But most of all these approaches used labeled dataset and cluster hate speech based on these labeled using different approaches with varying accuracy. In this approach we use feature extraction pair generation method for generating features. These features are then divided into three clusters positive, negative, and neutral. The negative cluster is further divided as "hate". "offensive" and clean". The summarization of our work is as follows:

1. We use semantic-based approach for extracting semantic feature and sentence form tweet comments.
2. Unigram features are extracted from sentence.
3. We use Sentiment Sensitive dictionary tagging in two stage:
    (a) Priority based part of speech dictionary tagging.
    (b) Priority dictionary tagging.
4. Opinion about particular category is described by direct stated feature extraction pair generation.
5. Summary generation represented through tag cloud for different scenario of feature opinion pair.

3 Related works

Based on the analysis of comments, tweets, posts on OSN, deep study and research applied on diverse area varying from sentiment analysis and classification (Soler et al. 2012; Homoceanu et al. 2011; Hodeghatta 2013) to detection of irony or identification of rumor (Zhao et al. 2015) etc. Nonetheless or fewer works have been deal with detection of hate speech ;Derczynski et al. 2013).

In Hajime Watanabe et al. (2018), perform hate speech detection based on the collected hateful and offensive expression and classify it in three class "Clean", "hateful", and "offensive". This classification is done based on pattern writing and feature extraction. Feature extraction further performs in four different ways. Firstly, feature extraction is done through classifying tweets into positive, negative, or neutral. This positive and negative score of words are extracted from SentiStrength, a tool that provide sentiment scores to words and sentences of which it is collected and composed. Secondly, semantic feature extraction is performed in which punctuations are extracted. They believed that punctuation marks helpful in detecting offensive for hateful speech. Third, unigram feature extracted from training set and perform PoS tagging of noun, verb, adjective or adverb and stored all in three different list one for each PoS tagging. If word is present in tweet then its corresponding feature value is set 'true', otherwise it is set to 'false'. During the extraction process they extracted total of 1373 words and unigram features of 1373 are defined. Fourth, in pattern extraction they divides the tweet word into two category—"Sentimental word (SW)", that labeled with tagged with PoS of noun, verb, adjective or adverb and "non-sentimental word (NSW)" that labeled with another PoS tagging. Pattern extracted from tweets in a way that words belongs to either "SW" or "NSW" are replaced by its corresponding simplified PoS tag. Finally they perform parameter optimization by providing different values to each parameter. Using these optimized parameter accuracy, recall, precision and F1-Score of classification is calculated using different classifier. Overall 2010 tweets are considered for experiments and provide accuracy of 78.4% accuracy for detecting whether the tweet is hate or not and an accuracy of 87.4% when detecting whether offensive tweets using binary classification (Fig. 1).

In Warner and Julia (2012), Warner et al. adapted hate-speech detection with word sense disambiguation. They used template based strategy to extract the features. These approach use set of 1000 paragraph and labeled each word in paragraph as literals, PoS tagging and Brown clusters. Each template is centered around single based unigram feature extraction and PoS tagging. For every paragraph all the possible templates were generated and maintained a count of each template based on their positive and negative occurrence. The log-odd are calculated as total value of the ratio of positive to negative occurrence. Template that do not recognizes as positive or negative is discarded as log-odds

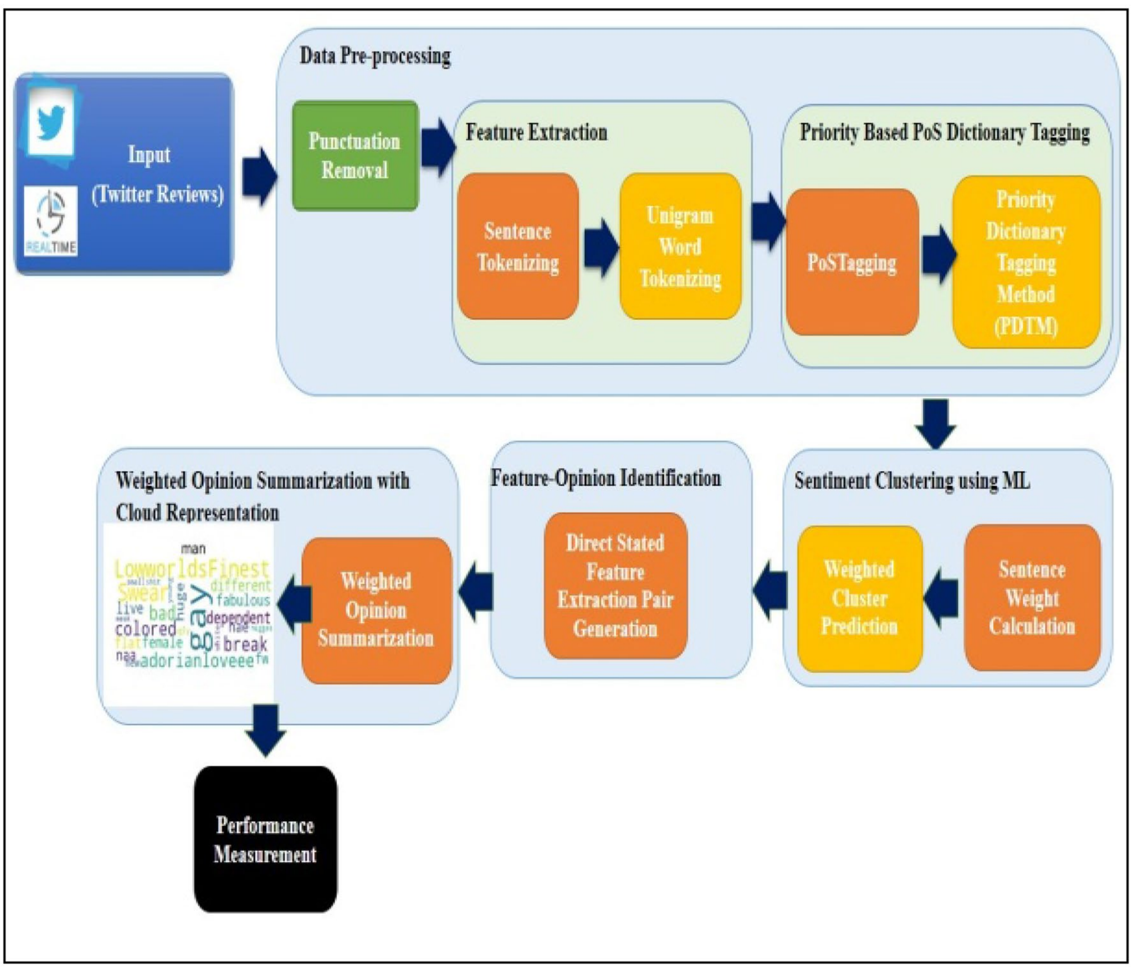

Fig. 1 Proposed framework

calculation is ratio based. Total 4379 features are processed. These extracted features are then classified into SVM classifier. This work provides accuracy of 94% of the classification using binary classification and F1 Score of 64%. The recall value of this system is low (Fig. 2).

In, Paula and Nunes (2018), describes current scenario, approaches, algorithms and unifying definition of hate speech and its detection. They analyze hate speech on different OSN websites, and contexts, based on the analysis they provide unifying definition of hate speech. Additionally represent several example and classification rules along with the argument that in favor or against of classification rule. With the conclusion of analysis they states that hate speech should compared for diverse area including discrimination, cyberbullying, toxicity, abusive language, flaming, radicalization and extremism. They specified that feature extraction of hate speech can be done by using general text mining algorithm including PoS based approach, sentiment analysis, n-gram approach, and deep learning. Hate speech feature extraction can also be done by with specific hate speech feature detection in order to address specified group, language or stereotype. Finally they showcase opportunities and challenges in this area, lack of tools and platform for automatic hate speech detection, insufficient comparative studies of existing approaches and lack of learning in languages excluding English (Fig. 3).

```
Input Sentence :  nice girls bad, make me get naughty. Bad yello hoe, real nice
body. Down south chick, like em real thick http://t.co/bzRDl3kF7U
-------------------------------------
Without Puncuation : nice girls bad make me get naughty Bad yello hoe real nice
body Down south chick like em real thick
-------------------------------------
Splitted Sentences :  [['nice', 'girls', 'bad', 'make', 'me', 'get', 'naughty',
'Bad', 'yello', 'hoe', 'real', 'nice', 'body', 'Down', 'south', 'chick', 'like',
 'em', 'real', 'thick']]
-------------------------------------
Tokanization :  [[('nice', 'nice', ['JJ']), ('girls', 'girls', ['NNS']), ('bad',
 'bad', ['JJ']), ('make', 'make', ['VBP']), ('me', 'me', ['PRP']), ('get', 'get'
, ['VB']), ('naughty', 'naughty', ['JJ']), ('Bad', 'Bad', ['NNP']), ('yello', 'y
ello', ['NNP']), ('hoe', 'hoe', ['NN']), ('real', 'real', ['JJ']), ('nice', 'nic
e', ['JJ']), ('body', 'body', ['NN']), ('Down', 'Down', ['NNP']), ('south', 'sou
th', ['RB']), ('chick', 'chick', ['VBP']), ('like', 'like', ['IN']), ('em', 'em'
, ['JJ']), ('real', 'real', ['JJ']), ('thick', 'thick', ['NN'])]]
-------------------------------------
Dictionary Tokanization [[('nice', 'nice', ['positive', 'JJ']), ('girls', 'girls
', ['Hate', 'NNS']), ('bad', 'bad', ['negative', 'JJ']), ('make', 'make', ['VBP'
]), ('me', 'me', ['PRP']), ('get', 'get', ['VB']), ('naughty', 'naughty', ['nega
tive', 'JJ']), ('bad', 'bad', ['negative', 'NNP']), ('yello', 'yello', ['NNP']),
 ('hoe', 'hoe', ['Hate', 'NN']), ('real', 'real', ['JJ']), ('nice', 'nice', ['po
sitive', 'JJ']), ('body', 'body', ['NN']), ('Down', 'Down', ['NNP']), ('south',
'south', ['RB']), ('chick', 'chick', ['VBP']), ('like', 'like', ['positive', 'IN
']), ('em', 'em', ['JJ']), ('real', 'real', ['JJ']), ('thick', 'thick', ['NN'])]
]
-------------------------------------
Hate Class
Hate Class
-4.0
```

Fig. 2 Stepwise Results

```
(girls,nice)
(girls,bad)
(girls,naughty)
(girls,real)
(girls,nice)
(girls,chick)
(girls,em)
(girls,real)
(girls,thick)
```

Fig. 3 Pair generation

In Gitari et al. (2015), the problem of hate speech is summarized into three fields of nationality, race and religion. Their work focuses on creating classifier for detecting hate speech on social media platform. The development of hate speech classifier and subjectivity analysis is being performed by rule-based learning approach. Objective sentence are then separated from subjective sentence using sentence—level sentiment analysis approach. Lexicon of hate speech is build up by extracting subjective and semantic word features. Lexicons are augmenting with noun pattern using bootstrapping. They annotated Lexicons of each sentence into three levels: "non-hateful (NH)", "strongly hateful (SH)" and "weekly hateful (SH)". Test the application over 500 labeled paragraphs. Using grammatic pattern and semantic feature run the classification application over labeled test set and provide F1-score of 65.12%.

In Olney et al. (2009), describes latent semantic analysis (LSA) generalization with the use local context and n-gram. Latent semantic analysis (LSA) is a technique of vector space which is used here to depict the sense of words. Usually, LSA carried out in two steps, in first step, the words are formed with the help of document matrix followed by the singular value decomposition of the matrix. However, document matrix is formed according to the word dimension and degree approximation of specified document. In this paper they use LSA to form content matrix which is characterized with feature rather than the word in document matrix (Fig. 4).

In Le and Mikolov (2014) address and work out over the detection of hate speech of online user comment. This is a two step approach in First step, joint modeling of word and comments is carried out using paragraph2vec. In second step, continuous bag-of-word (CBOW) neural language model is used represent the words and comments in joint space. after that, embeddings is used to training in which a

girl	good	4	
girl	pussy	3	
girl	bad	2	
girl	black	2	
girl	fat	2	
girl	fuck	2	
girl	correct	1	
Girl	mo	1	
Girl	nigga8221	1	

Fig. 4 Weight calculation for particular feature

game	dumb	1	
garrettwinker	dumb	5	
girls	dumb	15	
God	dumb	4	

Fig. 5 Opinion occurrence for feature

binary classifier classify comment into clean comment and hateful comments. The performance evaluation is carried out large scale data set of user comments collected from Finance website of Yahoo. This dataset consist of 56,280 hate speech comments and 895,456 clean comments. This work provides 94% accuracy of classification and 63.75% F1 score.

In Chikashi et al. (2016) targeted on abusive language detection of user content generated online. This approach perform classification task using n-gram features, lexicon features, syntactic features, pretrained features, linguistic features, "comment2vec" features and "word2vec" features. Based on feature and classification, user contents are classified into two class: "Clean" and "Abusive". This approach also used the same dataset used in Le and Mikolov (2014). This finance data of Yahoo consist of 56,280 comments which is labeled as "Abusive" and 895,456 comments are labeled "clean". The accuracy reaches to 90% and Recall, Precision and F-score are 79%, 77% and 78% respectively (Fig. 5).

However, few different works also carried out in this direction in which rather classifying comment into classes they targeted to specific group. With this approach in Kwok et al. (2013) perform detection hateful sentence in twitter against black people. In this work unigram features extraction is employed. The average accuracy of 76% is achieved for binary classification. Classifier is not trained for bigram feature so for some instance it gives erroneous result with an error rate of 24%. Focusing on the hate speech detection in the direction of a particular gender, racial group, or ethnicity are some of few areas over which research is still needed. As this work collect unigrams feature related to that particular group. Thus, the built-in-dictionary of these unigrams features cannot be use again for detecting hate speech towards another groups with the similar efficiency.

In Burnap and Williams (2015) used relationship between the word using typed together with bag of words (BoW) features extraction approach to make a distinction of hate speech expression from clean ones.

4 Proposed methodology

With a given set of tweet comment, our work performs clustering of tweet comment into three clusters positive, negative, and neutral. The negative cluster is further divided as "Hate". "Offensive" and Clean". Machine learning approach is used to for classification and extract the feature form labeled and unlabeled dataset.

4.1 Weighted pattern matching based data preprocessing

4.1.1 *Punctuation removal using regular expression R_E*

We represent each position of the weighted sequence as a vector that contains all the symbols of the alphabet and their corresponding probabilities; if a character does not appear in a specific position then its probability will be zero. As a result, the weighted matrix representation will be used for weighted sequences. At the first stage, input sentence have been sent to the punctuation removal module. To remove the punctuation we have used regular expression which removes irrelevant punctuation marks from the sentence. This module we have used at a beginning of approach because we are using real time twitter sentences which has hashtag(#), @, !!!,links and other punctuation marks.

$$\begin{aligned} R_E \;=\; & \text{“}\backslash w{+}{:}\backslash/\{2\}[\backslash d\backslash w-] + (\backslash.[\backslash d\backslash w-]+) \\ & *(?:(?:\backslash/[\hat{}\,\backslash s/]*)) \\ & *\text{','“} \;\; \ldots\; eq(1) \end{aligned} \tag{1}$$

E.g. if we have sentences like "!!!!“@selfiequeenbri: cause I'm tired of you big bitches coming for us skinny girls!!”"

Then after punctuation removal output sentence would be like "8220selfiequeenbri cause Im tired of you big bitches coming for us skinny girls8221".

4.1.2 Feature extraction

We perform the feature extraction using two phase tokenizing method.

(a) First phase: In this phase we performed sentence tokenizing which divide user's reviews in different sentences. This sentences splitting up has been done using PunktSentenceTokenizer from the nltk.tokenize.punkt module which normally used where we have huge data as an input and it tokenize the sentence with punctuation mark.

$$S_1 = \left\{S_{11}, S_{12}, \dots, S_{1n}\right\} \tag{2}$$

(b) Second phase: In this phase, each sentence have been subdivided into unigram words, therefore we have used TreebankWordTokenizer:

$$\sum_{i=1}^{n} T_i = \left\{W_1, W_2, \dots, W_n foreach S_{in} where S_{in} \in S\right\} \tag{3}$$

After tokenizing process output sentences will be tokenized "[['8220selfiequeenbri', 'cause', 'Im', 'tired', 'of', 'you', 'big', 'bitches', 'coming', 'for', 'us', 'skinny', 'girls8221']]".

4.1.3 Priority based part of speech dictionary tagging (PBPoSDT)

Here we used priority based dictionary tagging which works on a following priority rules:

- Lengthiest matches have higher precedence.
- Search is made from left to right.

4.1.3.1 Part-of-speech (PoS) tagging: (POST)

Now we perform Part-of-Speech(PoS) tagging in which we assigned tag to each word like N, V, ADJ, ADV, P, CON, DT, NN,JJ,PRO, NP, INT, mention in Eq. (4):

$$\sum P = [(Wi, Wi, [POSTag]) for (W, POSTag) in Si] \tag{4}$$

4.1.4 Priority dictionary tagging method: (PDTM)

This tagging performs sequential search from left to write which compare each word with a dictionary word to tokenize that word based on the categories. We have division for + ve, −ve, incrementive, decrementive, hate and offensive words. If there is an occurrence of any tag then it is added to the previous tagging. Word Comparison with each dictionary D_i $if Wi \in D_i where Di = \left\{D_1, D_2, \dots, D_n\right\}$

$$\sum [P, T_d] = \left(Wi, Wi, \left[DictTag, POSTag\right]\right) \tag{5}$$

e.g. In our case "bitches" word comes under hate division. Therefore its tagging would be ('bitches', 'bitches', ['Hate', 'NNS'])".

"tired " word will be considered as a negative word. So, it will add negative tagging like "('tired', 'tired', ['negative', 'VBD'])".

Similarly, ''skinny'' comes under negative polarity. So its tagging would be ('skinny', 'skinny', ['negative', 'VBP']).

4.1.5 Sentiment clustering using machine learning

Here we used machine learning approach for clustering based on sentiment. This approach is performed using sentence weight calculation and weight score prediction (Fig. 6).

4.1.5.1 Sentence weight calculation After this tagging we have calculated weight of entire sentence. To calculate weight, each token of above step will be assigned with some

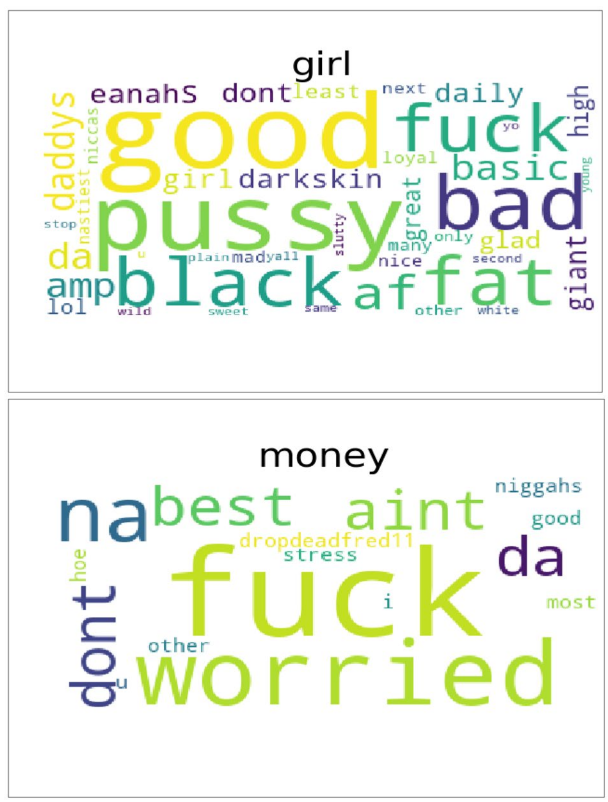

Fig. 6 Results of hate-offensive cluster based on tag cloud generation

weight. Positive is assigned with 1, negative polarity token has −1, offensive and hate is assigned with −2 weight-age. Normally, cumulative weight of all the word tokens is considered as a total weight of a sentence. But in many cases this gives wrong results. If in a sentence we have words like very, not, little then they will change the weight of a sentence. Words like very; amazingly, unbelievable gives additional weight to sentence. So for this type of incrementive word we will double the existing weight of a sentence. Words like barely, little reduces the impact of sentence. Therefore, we will calculate half of a existing weight. If we have word like not, then this will inverse the meaning of a sentence. So, for this type of words we have given a negation to a existing weight of a sentence.

If $\sum[P, T_d] has$ Positive

$$\sum_{i=1}^{n} SC = \{SC_1, SC_2, \ldots, SC_n\} = Score_{w_1} + Score_{w_2} + \cdots + Score_{w_n} \tag{6}$$

In our case, sentence has bitches, tired, skinny words. So, weight of sentences would be [bitches − hate − (−2) + tired − Negative − (−1) + skinny − Negative − (−1)] (−2 − 1 − 1 = −4).

4.1.6 Weighted cluster prediction

With a calculation of weight we can divide the sentence based on their weight. We have mainly positive, neutral, Negative cluster. But Negative cluster can be sub divided as Hate and offensive cluster.

If $\sum sc < 0$: negative cluster.

- If $H_t \in \sum[P, T_d]$: hate cluster.
- If $Offensive\_Tag \in \sum[P, T_d]$: offensive cluster.
- If$[Hate_Tag, Offensive_Tag] \notin \sum[P, T_d]$: clean negative cluster.

In our case total score we obtained is −4 for hate word therefore it belongs to hate cluster.

4.1.7 Performance measurement with the labeled dataset

Algorithm of proposed work

Step 1: Input Dataset

Step 2: For each sentence Si in S. Where $S = \{S_1, S_2, \ldots, S_n\}$

Step 3: Punctuation Removal using Regular Expression R_E

Where,R_E = “\w+:\/{2}[\d\w−] + (\.[\d\w−]+) ∗ (?:(?:\/[^\s/] ∗)) ∗ ’,’ “ … eq (7)

Step 4: Two Phase Tokenizing Process

- Does sentence tokenizing using PunktSentenceTokenizer
  $S_1 = \{S_{11}, S_{12}, \ldots, S_{1n}\}$
- Does word tokenizing using TreebankWordTokenizer

$$\sum_{i=1}^{n} T_i = \{W_1, W_2, \ldots, W_n \; for\; each\; S_{in} \; where\; S_{in} \in S\} \quad eq(8)$$

Step 5: Priority Based Part of Speech Dictionary Tagging

- POSTagging
  - Each ward assigned with Tag like N, V, ADJ , ADV , P, CON, DT, NN,JJ ,PRO, NP, INT.

$$\sum P = [(Wi, Wi, [POSTag])\; for\; (W, POSTag)\; in\; Si]$$

- Priority Dictionary Tagging
  - Word Comparison with each dictionary D_i $if\; Wi \in D_i\; where\; Di = \{D_1, D_2, \ldots, D_n\}$

$$\sum[P, T_d] = (Wi, Wi, [DictTag, POSTag])eq \tag{10}$$

Step 6: Sentence Weightage Calculation

If $\sum[P, T_d]$ *has* Positive

$$\sum_{i=1}^{n} SC = \{SC_1, SC_2, \ldots, SC_n\} = Score_{w_1} + Score_{w_2} + \cdots . + Score_{w_n}$$

Step 7: Weighted Cluster Prediction

- If $\sum sc < 0$: Negative Cluster
  - If $H_t \in \sum[P, T_d]$: Hate Cluster
  - If $Offensive\_Tag \in \sum[P, T_d]$: Offensive Cluster
  - If $[Hate\_Tag, Offensive\_Tag] \notin \sum[P, T_d]$: Clean Negative Cluster

Step 8: Performance measurement with the labeled dataset.

4.2 Feature opinion identification

Feature opinion identification is performed using direct stated feature extraction pair generation method (Fig. 7).

4.2.1 Direct stated feature extraction pair generation

Step 1: Input as a $\sum P$.

Step 2: Sentence which has directly stated opinion for particular feature that can be detected through noun-adjective pair.

- If $(NN \in \sum P(W_i) or NNP \in \sum P(W_i) or NNS \in \sum P(W_i) or NNPS \in \sum P(W_i))$

  Each $P(W_i)$ added to the NNDetect (NND) array.

- If $(JJ \in \sum P(W_i) or JJP \in \sum P(W_i) or JJS \in \sum P(W_i) or JJPS \in \sum P(W_i))$

  Each $P(W_i)$ added to the ADJDetect (ADJD) array.

Step 3: Calculation of length of NNDetect array and ADJD array.

Step 4: Pair generation $\sum W_i(NND, ADJD)$

- Case I—one NN one ADJPair (NND, ADJD).
- Case II—One NN More than one ADJPair((NND, ADJD1), (NND, ADJD2)…).
- Case III—More than One NN one ADJPair((NND1, ADJD), (NND2, ADJD)…).
- Case IV—More than One NN and ADJEach NND with its corresponding ADJ.

.

4.3 Weighted opinion summarization with cloud representation

Feature opinion pairs are used to summarize the opinion about particular object or person or some group of persons. We represent these feature opinion pair using tag cloud. Tag cloud represents information in pair wise manner. One element of pair object and other one is its feature. In tag cloud the information will reveal based on the following context:

- Which opinion has more weight or which Opinion is used more for particular feature.
- For example, For Girls which opinion has more weight? In our case, Good Opinion has highest weight.
- The particular opinion will occur frequently with which feature? In our case, for example dumb opinion is used maximum for girls.
- We can also generate the summary for other type of feature opinion pair e.g. Maximum occurrence of feature opinion pair, extracting the opinions of particular feature.

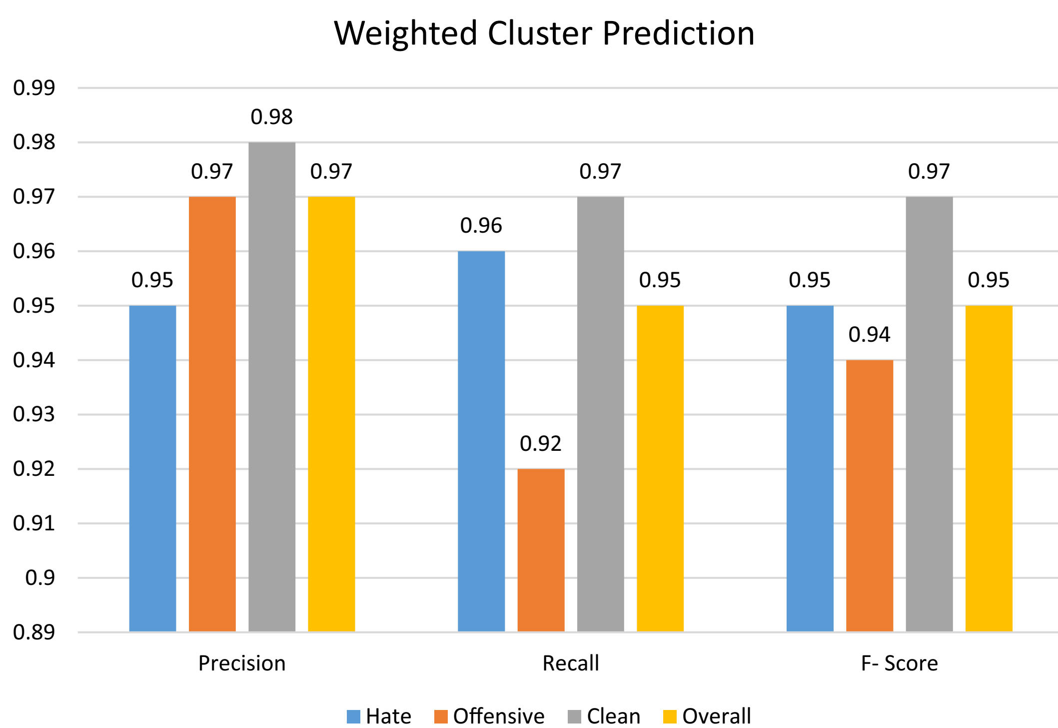

Fig. 7 Result of weighted class prediction on performance cluster at x-axis and accuracy at y-axis

For this type of user summary, we have to calculate the weight of feature opinion pair. This process has been done using grouping of feature opinion pair as whole one category and summation of each separate category has been done to calculate the weight. After calculation of frequency, representation of that weighted summary of each feature has been done. To represent a summary, word cloud representation has been performed for each unique feature. From this approach we can identify users' opinion for some particular category (Figs. 8, 9).

Step 1: Input $\sum W_i(NND, ADJD)$.

Step 2: Frequency calculation of $X_i = (NND_i, ADJ_i)$ as a one category of group.

$$Count_{X_i} = orcuurancesof(NNDi, ADJi)$$

$$\sum_{i-1}^{n} FC = \left\{ Count_{X_i} + Count_{X_2} + \cdots + Count_{X_n} \right\}$$

Step 3: For each feature $F_i$sort $(FC_{i1}, FC_{i2}, \ldots, FC_{in})$ for every opinion.

Step 4: Word cloud generation for every feature based on their weight. Which show higher font word higher weight.

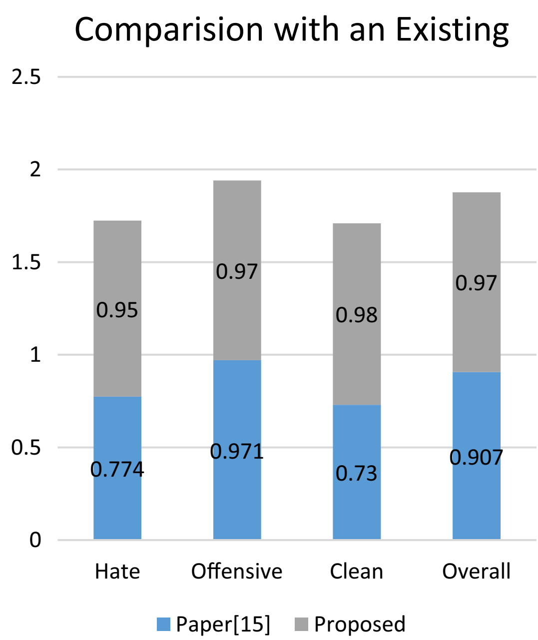

Fig. 8 Comparisons with a Paper Watanabe et al. (2018)

5 Results and discussion

5.1 Dataset description

The data set that we have collected is Twitter sentiment analysis, which is available online with free access. It comprises of around 1,00,000 tweets. The dataset link is https://www.kaggle.com/youben/twitter-sentiment-analysis/data (Fig. 10).

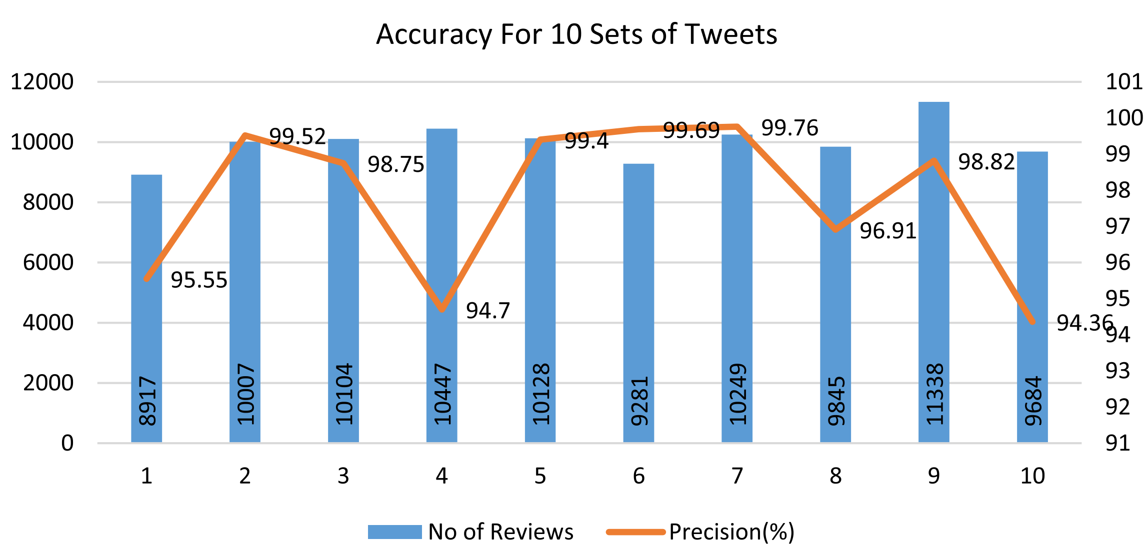

Fig. 9 Direct stated priority based pair generation method results no of sets at x- axis and no of tweets used at left y axis with accuracy at right y-axis

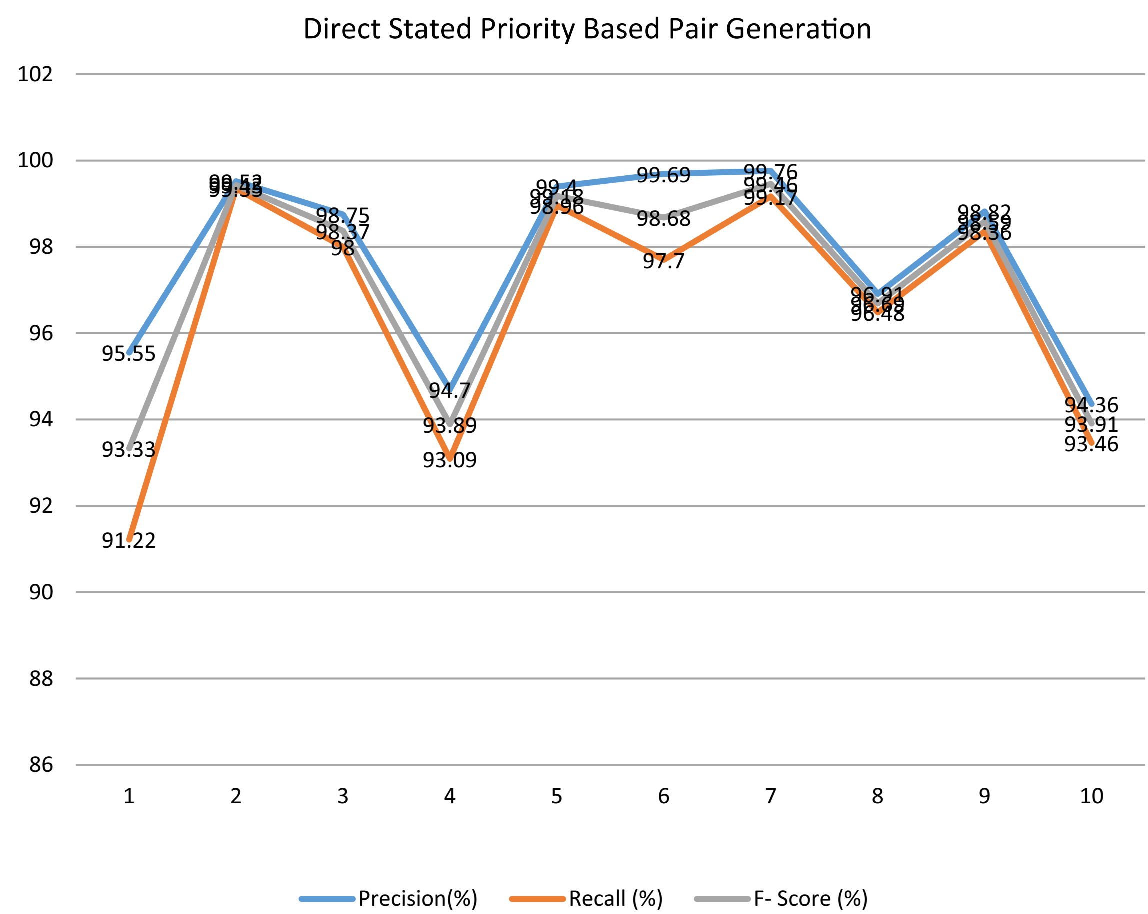

Fig. 10 Performance measurements for direct stated priority based pair generation method. Results no of sets at x-axis and accuracy at right y-axis

5.2 Experiment and result

From the dataset, we have taken 10 different sets of randomized comments for testing purpose. For performance measurements we have taken measurements of Precision based on correct identified to the total generated outcomes. Other than that, we have taken recall based on correct identified to the total input. To calculate the overall performance we have measured F-Score based on ratio of double of multiplication of precision and recall to the addition of precision and recall. At the first stage we have calculated the performance of cluster prediction based on priority. Refer Table 1 for result analysis of weighted cluster prediction which shows the performance measurements for hate, clean and offensive cluster. Graph shows that we have more than 90% of an accuracy in all parameters which is better than the existing approach. As mentioned in the Table 1 in base paper which have used same dataset has 0.907 overall accuracy that is 90% but in our approach we have accuracy 0.97 i.e. 97%.

In a second phase, we have performed the result analysis of direct stated priority based pair generation method. In which we have achieved overall 97.746% accuracy. In this approach, we have calculated the accuracies based on total no of comments in each stage with extracted and correct obtained pairs of those sentences (Tables 2, 3).

Table 1 Result analysis of clustering

Class	Precision	Recall	F-Score
Hate	0.95	0.96	0.95
Offensive	0.97	0.92	0.94
Clean	0.98	0.97	0.97
Overall	0.97	0.95	0.95

Table 2 Comparison with an existing approach

Cluster	Paper Watanabe et al. (2018)	Proposed
Hate	0.774	0.95
Offensive	0.971	0.97
Clean	0.73	0.98
Overall	0.907	0.97

6 Conclusion

The Hate speech is a form of thought, speech and social positioning that encourages violence against different groups in society. It can be verbalized or written and its intention is to discriminate against people because of their differences, be they race, color, ethnicity, religion, sexual orientation, disabilities etc. The Hate Speech is based on the hatred in itself of the different and all the prejudices that result from that feeling. This hate speech nowadays becomes very paper over online social networking website like "Facebook", Twitter" and "instagram". People feel physically feel safe and secure on this platform while exchanging their thoughts over the discussion forum provided by such websites. For any controversial topic, sharing of thoughts amongst people becomes very aggressive. It was found based on the survey report carried by Europe many times these hate crime are provoked by such activities. So detection of hate and offensive space on OSN website were found to be the demand of research. Earlier many researches were carried out that we had shown in related work section on this research area, in most cases they use labeled dataset and divide them indifferent hateful speech categories with varying categories. In this context we use sentiment sensitive dictionary based machine learning approach for hate speech detection on real time twitter tweets in which real time twitter are data are preprocessed using weighted pattern matching. For extracting semantic feature, punctuation removal based regular expression used. Two step tokenization phase extract sentence and unigram features from tweets. Priority based dictionary tagging is performed for calculating weight of each sentence is calculated and then based on weights of each in sentence tweets are clustered into Positive, negative, Hate offensive, and clean speech. The new feature that is added in our work is *Direct Stated Feature Extraction Pair Generation that provides stated opinion* for particular feature based on Noun-Adjective pair. Our approach represents tag cloud for different scenario of each feature pair generation. Our proposed approach has been tested on lakh tweets and then performed evaluation and validation over tested data that provide the average accuracy of 97% of the task using binary clusterization, F-Score is equal to 94%, and recall is 93%. We also provide comparative study of our work with existing approach. In future work, it is possible to predict or detect hate speech for tweets that are only opinion based in which object is absent for which opinion is given.

Table 3 Performance measurements of direct stated priority based pair generation method

Reviews	No of reviews	Extracted reviews	Correct obtained	Precision (%)	Recall (%)	F- Score (%)
Set 1	8917	8514	8135	95.55	91.22	93.33
Set 2	10,007	9991	9942	99.52	99.35	99.43
Set 3	10,104	10,027	9902	98.75	98	98.37
Set 4	10,447	10,269	9724	94.7	93.09	93.89
Set 5	10,128	10,083	10,023	99.4	98.96	99.18
Set 6	9281	9095	9067	99.69	97.7	98.68
Set 7	10,249	10,188	10,164	99.76	99.17	99.46
Set 8	9845	9801	9498	96.91	96.48	96.69
Set 9	11,338	11,286	11,153	98.82	98.36	98.59
Set 10	9684	9591	9051	94.36	93.46	93.91

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